

Server-Driven UI (SDUI) — Complete Resource Guide & Learning Plan

For Android Developers (Kotlin + Jetpack Compose)

Curated for Tarun Sharma — May 2026

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GitHub Repositories

★ Must-Study (Android / Kotlin)

Repo	Stars	What you'll learn
skydoves/server-driven-compose	★ ★ ★	Best starting point. Full SDUI with Jetpack Compose + Firebase Realtime DB. Covers rendering protocols, action handlers, versioning, and component design systems. Uses MVVM + Hilt + Retrofit. Perfect match for your AetheraAdmin stack.
arrazyfathan/server-driven-ui	★ ★	Concise, beginner-friendly SDUI example with Jetpack Compose. Good for understanding the core rendering loop before diving into larger projects.
tchigher/beagle	★ ★	Beagle — a full SDUI framework originally by Zup IT. Kotlin on both backend and client. Uses Moshi for deserialization. Learn how a production-grade SDUI framework is structured end-to-end.

Repo	Stars	What you'll learn
Kotlin SDUI with Hilt + Navigation (search topic)	—	Browse the <code>server-driven-ui</code> GitHub topic to discover fresh repos with Hilt, Navigation 3, and Compose Multiplatform implementations.

Cross-Platform / Framework Reference

Repo	What you'll learn
csmets/Server-Driven-UI	Full-stack SDUI framework with GraphQL compositor, template server, and React client. Teaches the backend side — how a server actually generates UI schemas. Essential for understanding the full picture.
MobileNativeFoundation/discussions#47	Not a repo but a goldmine — engineers from Airbnb, Lyft, Lyft, Wise, and startups share their real SDUI battle stories, code patterns, and architecture decisions.
nicklockwood/Euclid (iOS reference)	iOS SwiftUI SDUI reference if you ever need to go cross-platform with KMP.

Official Engineering Blogs

These are the **canonical real-world case studies** — read these like textbooks.

🏆 Tier 1 — Must Read (The Founding Papers)

1. Airbnb — "A Deep Dive into Airbnb's Server-Driven UI System"

- 🔗 <https://medium.com/airbnb-engineering/a-deep-dive-into-airbnbs-server-driven-ui-system-842244c5f5>
- Author: Ryan Brooks
- This is *the* canonical paper for SDUI in mobile. Covers their Ghost Platform, how they serve Android + iOS + Web from the same schema, component versioning, and A/B testing at scale. Every SDUI system traces back to this.

2. Swiggy — "A Deep Dive into Dynamic Widget: Swiggy's Server-Driven UI System"

- 🔗 <https://bytes.swiggy.com/a-deep-dive-into-dynamic-widget-swiggys-server-driven-ui-system-92cdc3b16ec6>

- This is exactly the system you noticed in the Swiggy app. Covers how Swiggy uses Litho + SDUI for the home feed, location-based widgets, and dynamic banners. Real Indian context.

3. Netflix — "Server-Driven UI for Mobile and Beyond" (QCon London 2024)

- <https://www.infoq.com/presentations/server-ui-mobile/>
- Staff Engineer Christopher Luu explains Netflix's UMA (Universal Messaging Alert) system, the spectrum of SDUI, backwards compatibility via GraphQL, and how they extended SDUI to TV + Web beyond just mobile.

4. Lyft — "Canvas: A Server-Driven Framework for Mobile"

- <https://eng.lyft.com/canvas-a-new-way-for-lyft-to-describe-mobile-uis-e2de28d6bb88>
- Lyft's Canvas uses Protocol Buffers instead of JSON for a compact binary format. Learn why they chose protobuf, how they define primitives (buttons, layouts, action callbacks), and their versioning strategy.

🏆 Tier 2 — Company-Level Case Studies

5. Reddit — Adopts SDUI for New Feed Architecture

- <https://www.infoq.com/news/2023/09/reddit-feed-server-driven-ui/>
- Reddit rebuilt their 6-year-old feed codebase (originally Objective-C) using SDUI. Result: home feed 12% faster to load. Learn how they reworked their GraphQL API response structure.

6. DoorDash — "Mosaic: Server-Driven UI"

- [Search "DoorDash Mosaic server driven UI engineering blog"](#)
- DoorDash's Mosaic system — reduces time to deliver new banners and tags to under a day, modifications in less than an hour.

7. Doist (Todoist) — "Server-Driven UI from a Mobile Perspective"

- <https://doist.dev/posts/server-driven-ui-from-a-mobile-perspective>
- Excellent balanced perspective from a smaller company. Covers the tradeoffs honestly, how they built their own SDUI SDK, and platform parity across Android + iOS.

8. Q42 — "One Code to Rule Them All: Android Server Driven UI"

- <https://engineering.q42.nl/android-server-driven-ui/>
- Deep Android-focused case study. Uses Kotlin sealed classes as the type hierarchy, covers the full data → business logic → render flow. Explains how even local DB sync

works differently with SDUI.

9. Mercari — "Server-Driven UI for Marketing Campaigns"

- 🔗 Search "Mercari server driven UI marketing engineering blog"
- How Mercari implemented SDUI specifically for time-limited campaigns — remote UI configuration with native performance.

10. Zalando — "Evolving SDUI Framework for Scale"

- 🔗 Search "Zalando server driven UI framework"
- European e-commerce giant's lessons on scaling SDUI to dozens of teams.

11. Flipkart — Proteus (Their Own SDUI Engine)

- 🔗 <https://proandroiddev.com/dynamic-screens-using-server-driven-ui-in-android-262f1e7875c1>
- Flipkart built *Proteus* — a drop-in replacement for Android's LayoutInflater that inflates JSON layouts at runtime. This is the Indian equivalent of Airbnb's Ghost Platform.
- GitHub: <https://github.com/flipkart-incubator/proteus>

Articles & Blog Posts

Foundation Articles (Start Here)

Article	Link	Why Read
Understanding Client-Driven UI Limitations & Rise of SDUI	https://medium.com/androididiots/understanding-client-driven-ui-limitations-and-the-rise-of-server-driven-ui-372a71c9c362	Great conceptual intro — explains <i>why</i> SDUI, not just <i>what</i>
Server-Driven UI Android Implementation — Shubham Agrawal	https://medium.com/@iagrawalshubham/server-driven-ui-android-implementation-e4ae865b10d0	Step-by-step Android implementation with code
JetPack Compose With Server Driven UI — Siva Ganesh	https://medium.com/android-dev-hacks/jetpack-compose-with-server-driven-ui-396a19f0a661	Compose-specific SDUI patterns, reactive approach
Server-Driven UI in Android with	https://medium.com/basalam/server-driven-ui-in-android-with-compose-bf1885e9343c	Real-world multi-part series from a

Article	Link	Why Read
Compose (Part 1) — Basalam		production team
Understanding SDUI with Android and Jetpack Compose — Dhiraj Thakur	https://medium.com/@dhirajkumar.dt51/understanding-server-driven-ui-with-android-and-jetpack-compose-ec1427b60ccd	Feb 2025, recent + practical with Firebase
Android SDUI — XML vs Compose Benchmark — İbrahim Ethem Şen	https://medium.com/@ibrahimethemsen/android-server-driven-ui-xml-vs-compose-example-benchmark-827a71d6605b	Side-by-side comparison + performance benchmarks

Intermediate / Architecture Deep Dives

Article	Link	Why Read
Server Driven UI — Concepts and Building Blocks — Comviva	https://medium.com/@dfs.techblog/server-driven-ui-concept-db07d7946e94	Strong conceptual foundation for the building blocks
Server Driven UI for Mobile Apps — Qantas Engineering	https://medium.com/qantas-engineering-blog/server-driven-ui-for-mobile-apps-48f8488ed7a4	Pros/cons + backend design from an airline super app
SDUI: The Necessary Evil for Scalable Mobile Apps — Tushar Gupta	https://medium.com/digia-studio/server-driven-ui-sd-ui-the-necessary-evil-for-scalable-mobile-apps-80c650a2c8de	Excellent Dec 2025 article covering why every growth-stage company ends up here
Server Driven UI — RapiPay Engineering	https://medium.com/@tech.rapipay/server-driven-ui-80ae85603747	Fintech perspective — strict data + UI separation
Dynamic Screens Using SDUI in Android — ProAndroidDev	https://proandroiddev.com/dynamic-screens-using-server-driven-ui-in-android-262f1e7875c1	Covers Swiggy (Litho), Flipkart (Proteus), Airbnb (Epoxy) — Indian app ecosystem focus

Article	Link	Why Read
Design SDUI with Jetpack Compose and Firebase — Stream/GetStream	https://getstream.io/blog/server-driven-compose-firebase/	Deep technical blog post accompanying the skydoves repo. Covers layout nodes, action handlers, and versioning in detail.
Case Study: Swiggy's Dynamic Widgets & SDUI	https://medium.com/@viditsavaliya/cash-study-swiggy-dynamic-widgets-server-driven-ui-a-deep-dive-44adf0801c1d	Swiggy-specific deep dive with Jetpack Compose code snippets
What Airbnb, Netflix, and Lyft Learned	https://medium.com/@aubreyhaskett/server-driven-ui-what-airbnb-netflix-and-lyft-learned-building-dynamic-mobile-experiences-20e346265305	Dec 2025 synthesis of all three companies' lessons
Server Driven UI — Dev.to	https://dev.to/nishant_keshav/server-driven-ui-3l0p	Good community-level explanation, beginner-friendly

YouTube Videos & Conference Talks

Direct YouTube Videos

Video	Channel	Why Watch
How to Build Server Driven UI w/ Firebase + Jetpack Compose	YouTube	 https://www.youtube.com/watch?v=tca-6yhWXNo — Hands-on tutorial, directly builds the system. Best for learning-by-coding alongside it.
Server Driven UI with Jetpack Compose	Curated Reality	 https://www.youtube.com/watch?v=rE9ZEODzLFs — Compose-specific walkthrough
A Page Out of Server Driven UI on Android — Adit Lal	Droidcon	 https://www.youtube.com/watch?v=caX3GLXXGmw — Conference talk: emphasizes why SDUI is a hot topic and how to architect it
Server Driven UI, Tom Lokhorst	CocoaHeads	 https://www.youtube.com/watch?v=ERPmUsLkwEE — Real production implementation walkthrough

Conference Talks (Video + Transcript)

Talk	Conference	Link
Server-Driven UI for Mobile and Beyond — Christopher Luu (Netflix)	QCon London 2024	https://www.infoq.com/presentations/server-ui-mobile/
Server Driven UI — Streamlining Mobile Dev and Release — Thomas Chao	QCon	https://www.infoq.com/presentations/sduie/
Dynamic Flow: Wise's KMP Approach to SDUI	Droidcon 2024	https://www.droidcon.com/2024/11/22/dynamic-flow-the-wise-approach-to-server-driven-ui/

Search These Terms on YouTube

- "server driven UI" android jetpack compose
- "SDUI" android kotlin tutorial
- droidcon server driven UI
- "backend driven UI" android
- Flipkart Proteus Android



Documentation & Reference Sites

Resource	Link	What
Jetpack Compose.app — SDUI Libraries	https://www.jetpackcompose.app/Server-Driven-UI-libraries-in-Jetpack-Compose	Curat SDUI code : Jetpa
GitHub Topic: server-driven-ui	https://github.com/topics/server-driven-ui	All pu tagge topic, stars. and c week

Resource	Link	What
MobileNativeFoundation Discussions	https://github.com/MobileNativeFoundation/discussions/discussions/47	The b comn discu SDUI acros and p
Flipkart Proteus GitHub + Docs	https://github.com/flipkart-incubator/proteus	India: engin Andro docu JSON exam
Beagle Framework Docs	https://docs.usebeagle.io	Full S frame Zup I backe gener Andro rende gettir guide
Mobile Vitals — Android SDUI	https://mobile-vitals.com/platform/android	Aggre engin posts major Filter Drive
InfoQ Mobile Track	https://www.infoq.com/mobile/	Regul publi: case s QCor confe

Podcasts

Podcast	Episode	Link
Lyft Mobile Podcast	"Server Driven UI with Kevin Fang and Jeff Hurray"	Search Lyft Engineering podcast — discusses their Canvas SDUI system
Android Developers Backstage	Various Jetpack Compose episodes	https://adbackstage.libsyn.com/
Fragmented Podcast	Episodes on modern Android architecture	https://fragmentedpodcast.com/

Other Resources

Tools & Libraries to Know

Tool/Library	Purpose	Link
Proteus (Flipkart)	Drop-in LayoutInflater replacement, inflates JSON at runtime	github.com/flipkart-incubator/proteus
Litho (Meta)	Declarative UI for complex Android views — Swiggy uses this	fblitho.com
Epoxy (Airbnb)	RecyclerView + SDUI for traditional Views	github.com/airbnb/epoxy
Moshi	JSON deserialization — better than Gson for sealed class polymorphism	github.com/square/moshi
Kotlinx.serialization	Official Kotlin JSON parser with sealed class support	kotlinlang.org/docs/serialization
Beagle	Full SDUI framework (backend + mobile) by Zup IT	docs.usebeagle.io
Compose Multiplatform	Take your SDUI renderer to iOS + Desktop	jetbrains.com/lp/compose-multiplatform

Communities

- **Android Dev Discord** — #architecture channel
 - **Kotlin Slack** — #compose channel
 - **r/androiddev** — search "SDUI" or "server driven"
 - **ProAndroidDev** publication on Medium
-

12-Week Learning & Implementation Plan

Goal: By Week 12, you will have built a working SDUI module in your **Aethera** e-commerce app and can confidently discuss it in interviews.

Phase 1 — Foundation (Weeks 1-2)

Understand the concept deeply before writing a single line.

Week 1 — The "Why"

- ☐ Read: "Understanding Client-Driven UI Limitations & Rise of SDUI" (Medium/AndroIDIOTS)
- ☐ Read: Airbnb Ghost Platform blog post (the canonical paper)
- ☐ Read: Swiggy's Dynamic Widget blog post (because it's literally what you noticed)
- ☐ Read: Doist's "SDUI from a Mobile Perspective" (honest tradeoffs)
- ☐ Watch: "A Page Out of Server Driven UI on Android" — Adit Lal (Droidcon YouTube)

Deliverable: Write 1 page of notes answering — "What problem does SDUI solve? When would I use it in Aethera?"

Week 2 — The Architecture

- ☐ Read: Q42's "One Code to Rule Them All: Android Server Driven UI"
- ☐ Read: "Server Driven UI — Concepts and Building Blocks" (Comviva)
- ☐ Study: MobileNativeFoundation Discussion #47 (at least the first 15 comments)
- ☐ Watch: QCon talk by Christopher Luu (Netflix) on InfoQ

Deliverable: Draw your own diagram of the SDUI data flow for Aethera — from server JSON → Kotlin sealed class → Composable

Phase 2 — First Implementation (Weeks 3-5)

Get hands dirty with a real repo.

Week 3 — Clone & Explore

- ☐ Clone `skydoves/server-driven-compose` — run it on your physical device
- ☐ Read the GetStream blog post alongside the repo code
- ☐ Map every component: Where is the JSON parsed? Where is `when(component)` written? Where are Composables called?
- ☐ Read the `server-driven-compose` data model files — understand the sealed class hierarchy

Week 4 — Build Your First SDUI Screen

- ☐ Create a new branch in `AetheraAdmin`: `feature/sd-ui-home-feed`
- ☐ Define your first 3 component types as sealed classes: `BannerCarousel`, `CategoryStrip`, `ProductRow`
- ☐ Write the JSON schema for each component (create mock JSON files)
- ☐ Write the Moshi/Kotlinx deserializer with polymorphism
- ☐ Write the `SdUiRenderer` composable with `when(component)` dispatch

Week 5 — Wire to Real Data

- ☐ Create a mock API endpoint (can be a local JSON file served by Ktor or just hardcoded for now)
- ☐ Build the `SdUiViewModel` with `StateFlow` + `Loading/Success/Error` states
- ☐ Handle the `Unknown` component type gracefully (no crash on unknown types)
- ☐ Add shimmer/skeleton loading while components load

Deliverable: Working SDUI home feed with 3 component types rendering from JSON

Phase 3 — Production Patterns (Weeks 6-8)

Level up to real-world requirements.

Week 6 — Actions & Navigation

- ☐ Design your action schema: `deeplink`, `navigate`, `api_call`, `bottom_sheet`
- ☐ Implement `SdUiAction` sealed class
- ☐ Wire button taps → action handler → `NavController` / deep link router
- ☐ Study: How Lyft's Canvas handles action callbacks in protobuf
- ☐ Read: DoorDash "Facets" framework blog post

Week 7 — Versioning & Backwards Compatibility

- ☐ Study how `skydoves/server-driven-compose` handles versioning
- ☐ Implement a `minVersion` / `maxVersion` field on components
- ☐ Add logic: if app version < component's `minVersion`, skip component silently
- ☐ Read: Netflix's GraphQL backwards-compatibility strategy (from QCon talk)

- ☐ Implement: Server sends `app_version` as query param, backend responds with compatible component set

Week 8 — Caching & Performance

- ☐ Read: Swiggy's caching mechanism section in their blog post
- ☐ Implement: Cache last successful SDUI response in DataStore / Room
- ☐ Show cached UI instantly on app open → refresh in background (stale-while-revalidate pattern)
- ☐ Measure: Compare first-paint time with vs without caching using Android Profiler

Deliverable: SDUI screen that loads from cache instantly and silently refreshes

Phase 4 — Advanced & Showcase (Weeks 9-11)

Build your flagship SDUI feature.

Week 9 — Theming & Design Tokens

- ☐ Add color/style tokens to your JSON schema (`backgroundColor`, `textColor`, `cornerRadius`)
- ☐ Wire tokens to your Composables via `LocalCompositionLocal` or direct mapping
- ☐ Test: Change a component's color in JSON → verify it changes on device without app update
- ☐ Study: AirAsia's design token adoption blog post (Mobile Vitals aggregator)

Week 10 — A/B Testing Simulation

- ☐ Create two variants of your home feed JSON (Layout A: banner first, Layout B: categories first)
- ☐ Add a `user_segment` field to your API (based on user ID % 2)
- ☐ Server returns different component order based on segment
- ☐ Add analytics events to each component render (can be simple logs for now)
- ☐ Read: Shopify's "launch experiments whenever we deem necessary" blog post

Week 11 — SDUI for Aethera Customer App

- ☐ Apply all learnings to the *customer-facing* Aethera app (not just admin)
 - ☐ Implement full SDUI home screen: banner carousel + category strip + product rows
 - ☐ Write a 500-word technical README explaining your SDUI implementation — this goes on your portfolio
 - ☐ Record a 3-minute screen recording demo for your portfolio website
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Phase 5 — Interview Readiness (Week 12)

Week 12 — Consolidate & Tell the Story

- ☐ Prepare 5 interview answers:
 - "What is SDUI and why would you use it?" (2 min answer)
 - "How would you implement SDUI in Jetpack Compose?" (3 min with code walkthrough)
 - "How do you handle backwards compatibility in SDUI?" (2 min)
 - "What are the tradeoffs of SDUI?" (2 min — be honest, show maturity)
 - "Give me an example of a company that uses SDUI and how they do it." (name Swiggy/Zepto — your discovery story is a great hook)
- ☐ Add "Server-Driven UI" to your resume skills and AetheraAdmin project description:

"Implemented a Server-Driven UI module enabling the home feed layout and component order to be controlled server-side without app updates, using sealed classes, Moshi polymorphic deserialization, and Jetpack Compose rendering dispatch."
- ☐ Push the SDUI branch to your AetheraAdmin GitHub repo
- ☐ Add SDUI section to your portfolio website
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Quick Reference: Learning Order by Week

Week 1-2	→ Read (Theory & Company Case Studies)
Week 3	→ Explore (Clone & Study Real Code)
Week 4-5	→ Build (First Working Implementation)
Week 6-7	→ Level Up (Actions, Versioning)
Week 8	→ Optimize (Caching, Performance)
Week 9-10	→ Advanced (Theming, A/B Testing)
Week 11	→ Ship (Portfolio-Ready Feature)
Week 12	→ Interview Prep

Starred Resources for Quick Revisit

#	Resource	Type	Priority
1	Airbnb Ghost Platform Blog	Article	Critical
2	skydoves/server-driven-compose	GitHub	Critical
3	Swiggy Dynamic Widget Blog	Article	Critical
4	GetStream SDUI + Firebase Blog	Article	High
5	Q42 Android SDUI Engineering Blog	Article	High

#	Resource	Type	Priority
6	Netflix QCon 2024 Talk	Video/Transcript	 High
7	MobileNativeFoundation Discussion #47	Community	 High
8	Flipkart Proteus GitHub	Git Hub	 Medium
9	Doist SDUI Blog	Article	 Medium
10	Lyft Canvas Engineering Post	Article	 Medium

Happy building, Tarun. The fact that you noticed SDUI in production apps before studying it means you're already thinking like an engineer, not just a developer.